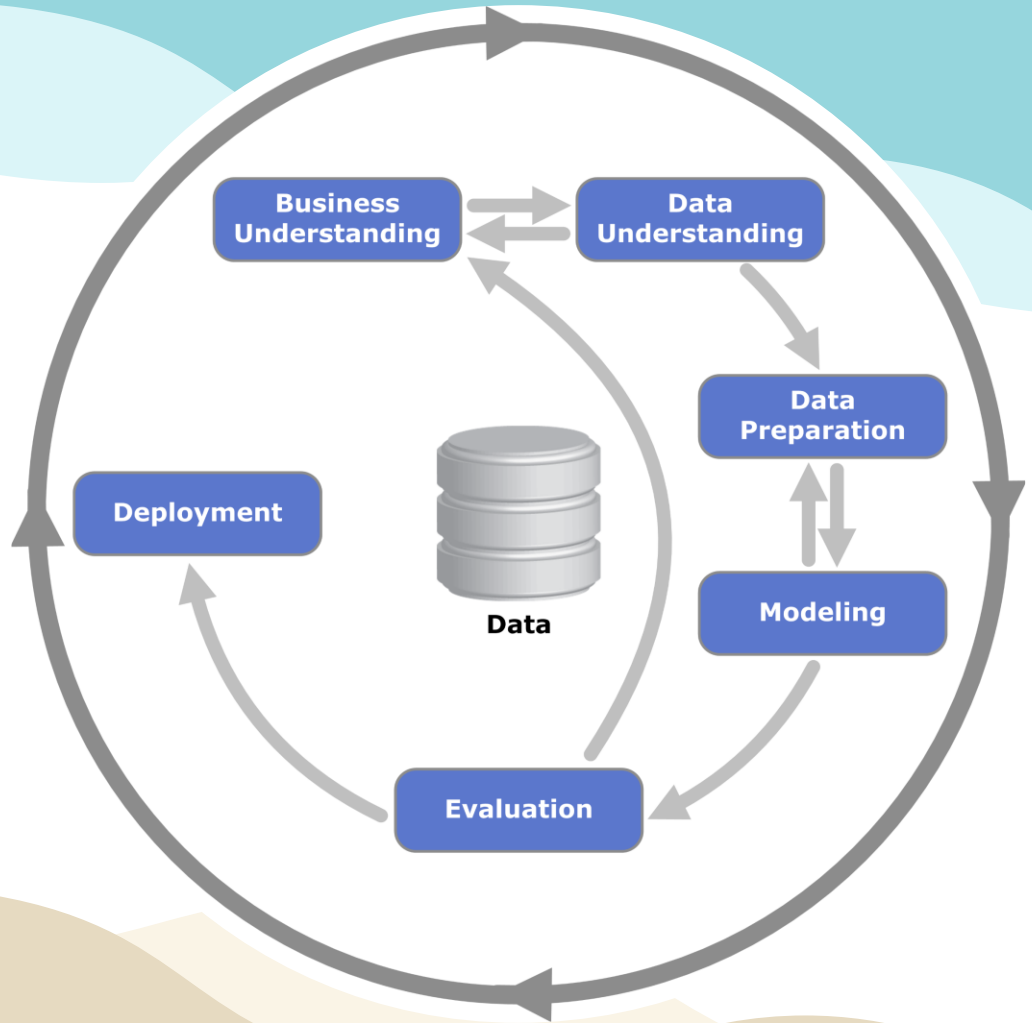


My Hero's Journey

Automated Resistor Value Recognition



The Cross-Industry Standard Process for Data Mining (CRISP-DM)

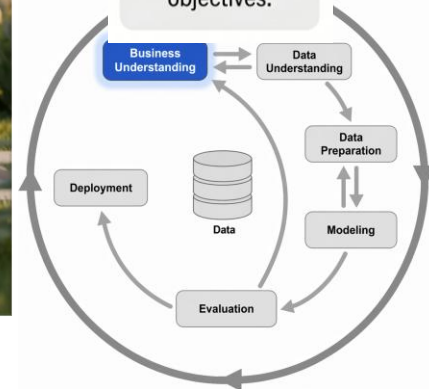


Business Understanding

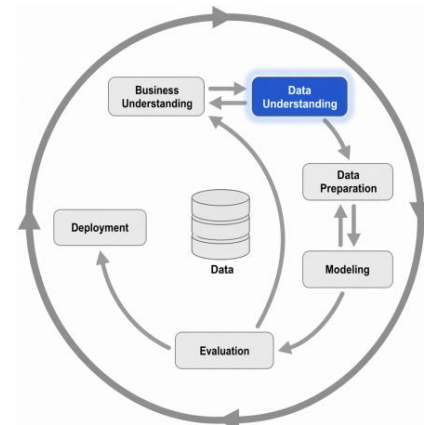
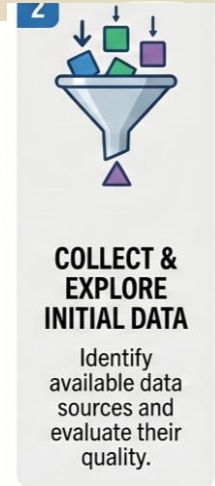


DEFINE BUSINESS GOALS & REQUIREMENTS

Align the data project with specific organizational objectives.



Data Understanding

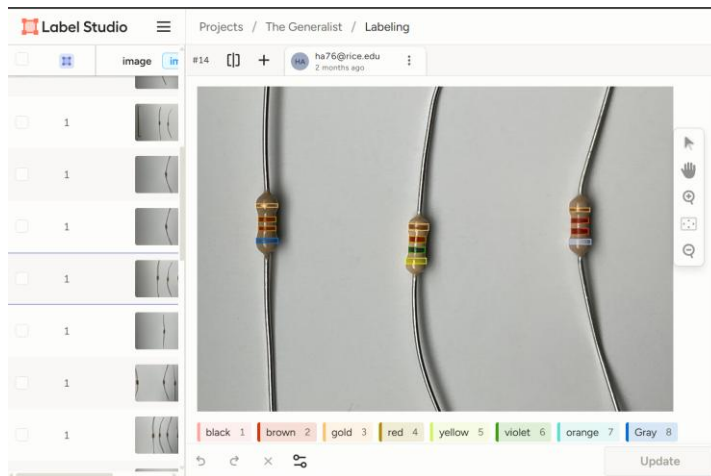
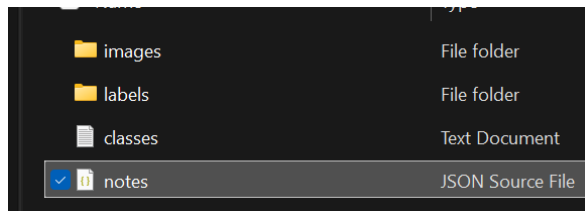
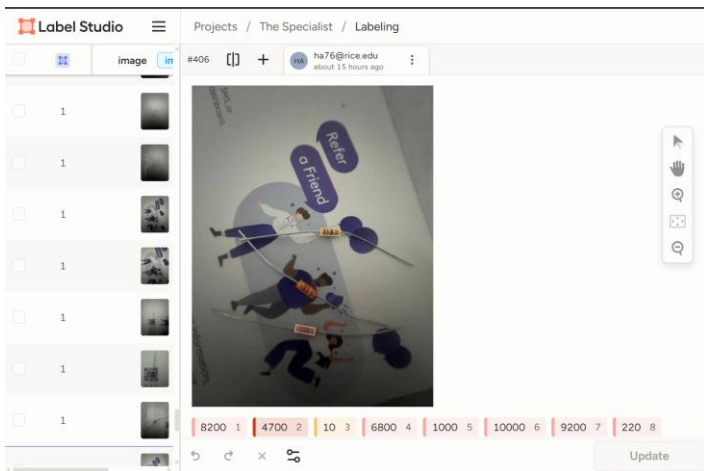


Data Preparation

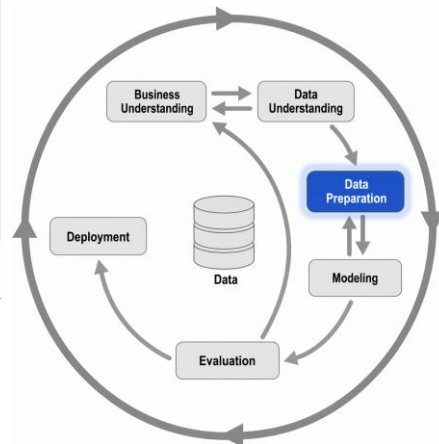


CLEAN & FORMAT DATA FOR ANALYSIS

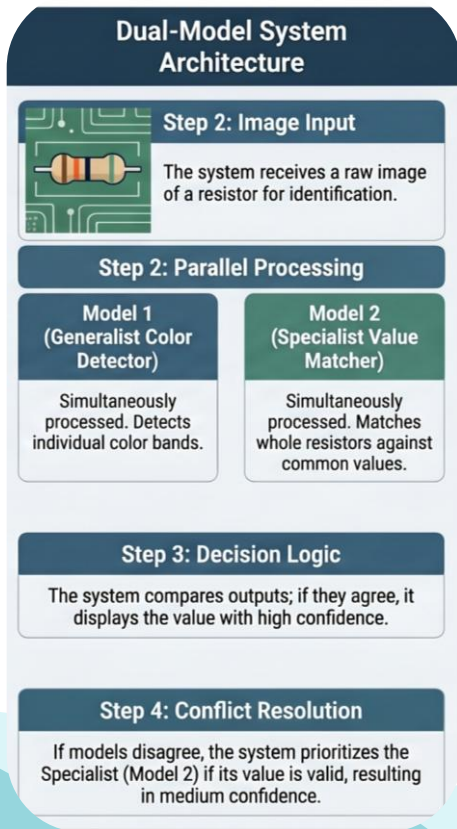
Transform raw datasets into a final format ready for modeling.



Label Studio



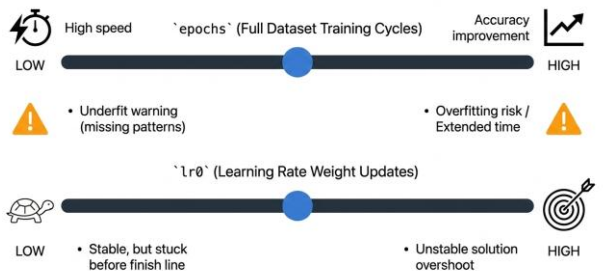
Modeling



Hardware & Structural Hyperparameters

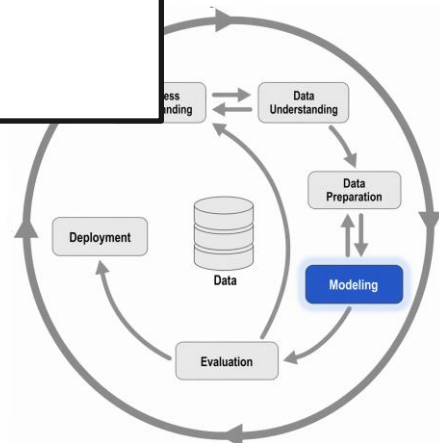
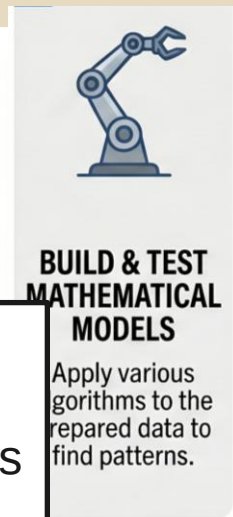


Learning Mechanics & Convergence Hyperparameters



Final
tuned
hyperparameters
will be
presented
here

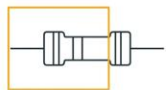
```
epochs=40 \
imgsz=640 \
batch=16 \
lr0=0.005
```



Evaluation

Training Diagnostics Demystify Error Types

`'box_loss'` (Location Error)



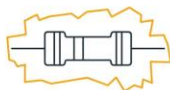
HIGH: Poor localization; model completely misses object placement.
LOW: Accurate localization.

`'cls_loss'` (Class Error)



HIGH: High rate of misclassifications despite finding the object.
LOW: Good class predictions.

`'dfl_loss'` (Edge Precision Error)



HIGH: Inaccurate, loose, or jagged boundary definitions.
LOW: Precise, tight bounding boxes.

PyTorchbookLM

Defining Success Through the Precision-Recall Tradeoff

Precision (Box P): The Accuracy of Guesses



When the model predicts, is it correct?

HIGH Precision = Few false positives. The model is accurate when it commits to a prediction.

Recall (R): The Completeness of Vision



Did the model find all the real objects?

HIGH Recall = Finds most objects, few misses. The model successfully detects what is actually there.

PyTorchbookLM



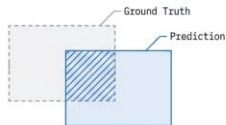
**EVALUATE
RESULTS
AGAINST
BUSINESS
GOALS**

Verify that the model's findings actually solve the original problem.

Strict Accuracy Standards Require High Overlap

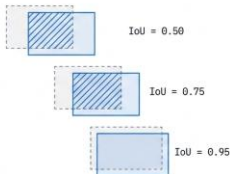
Intersection over Union (IoU) Defines mAP

`'mAP50'`



Standard overall detection accuracy at an IoU threshold of 0.5. HIGH indicates strong baseline performance.

`'mAP50-95'`



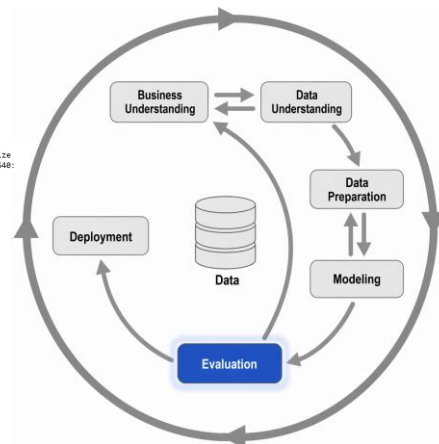
Strict accuracy graded across multiple escalating IoU thresholds. HIGH proves the model is exceptionally reliable, robust, and generalizes perfectly.

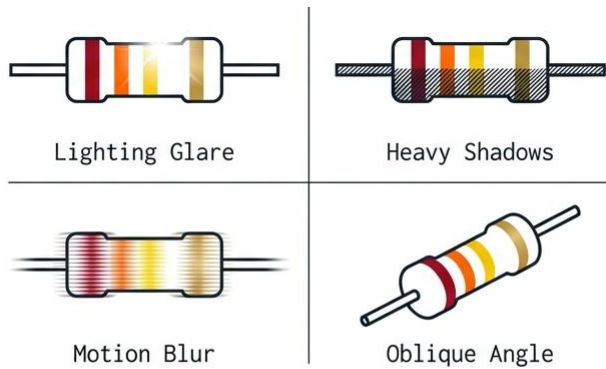
PyTorchbookLM

Final
evaluation
will be
presented
here

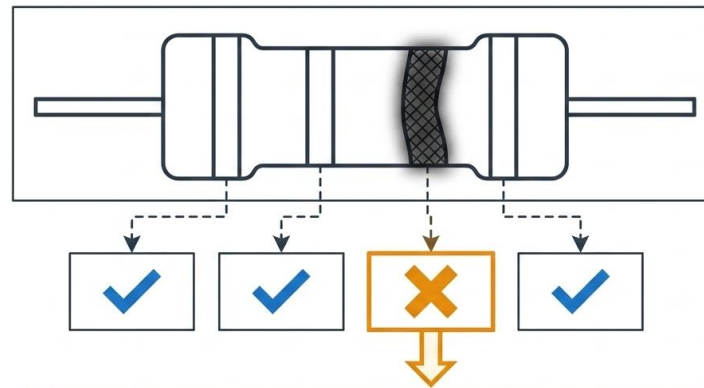
Class	Images	Instances	Box(P)	R	mAP50	mAP50-95
411	59	67	0.972	0.993	0.986	0.921
10	7	7	0.987	0.957	0.925	0.567
1000	7	7	0.973	1	0.995	0.876
10000	10	10	0.981	1	0.995	0.892
20000	12	12	0.984	1	0.995	0.953
210	6	6	0.952	1	0.995	0.919
310	5	5	0.972	1	0.995	0.945
4700	4	4	0.979	1	0.995	0.771
6000	6	6	0.969	1	0.995	0.818
8100	8	8	1	0.669	0.971	0.686
9200	2	2	0.925	1	0.995	0.846

Epoch 40/40 GPU_mem 4.87G box_loss 0.3788 cls_loss 0.4145 dfl_loss 0.8321 Instances 11 Size 640:



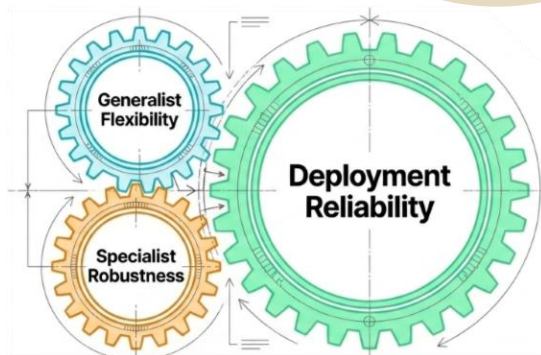


Real-world variations cause fatal misclassification of individual bands.



ERROR: INCORRECT COMPUTED VALUE

Deployment?



DEPLOY INSIGHTS INTO PRODUCTION

Move the final findings or models into a live environment for use.

Navigation

Select AI Logic:

- ☒ Specialist Model
- ☐ Generalist Model (Coming Soon)
- ☐ Smart Logic (Coming Soon)

Session Inventory

Value	Confidence
9.2 k Ω	86.0%

Download CSV

Clear Inventory

Analysis

> View Image Quality Diagnostics



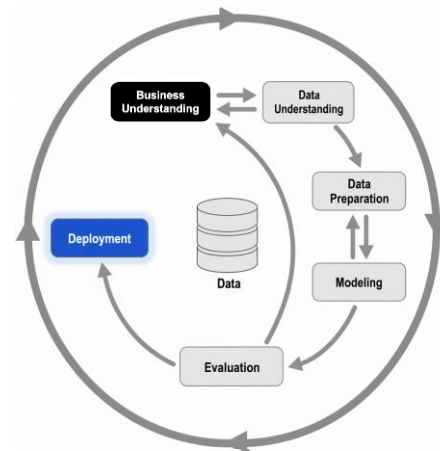
AI Detection Result

Detected Values

4.7 k Ω

Confidence: 88.2%

Save to Inventory



<https://resistor-detector-fcshdi9kgsW59ceu5bmpo.streamlit.app/>

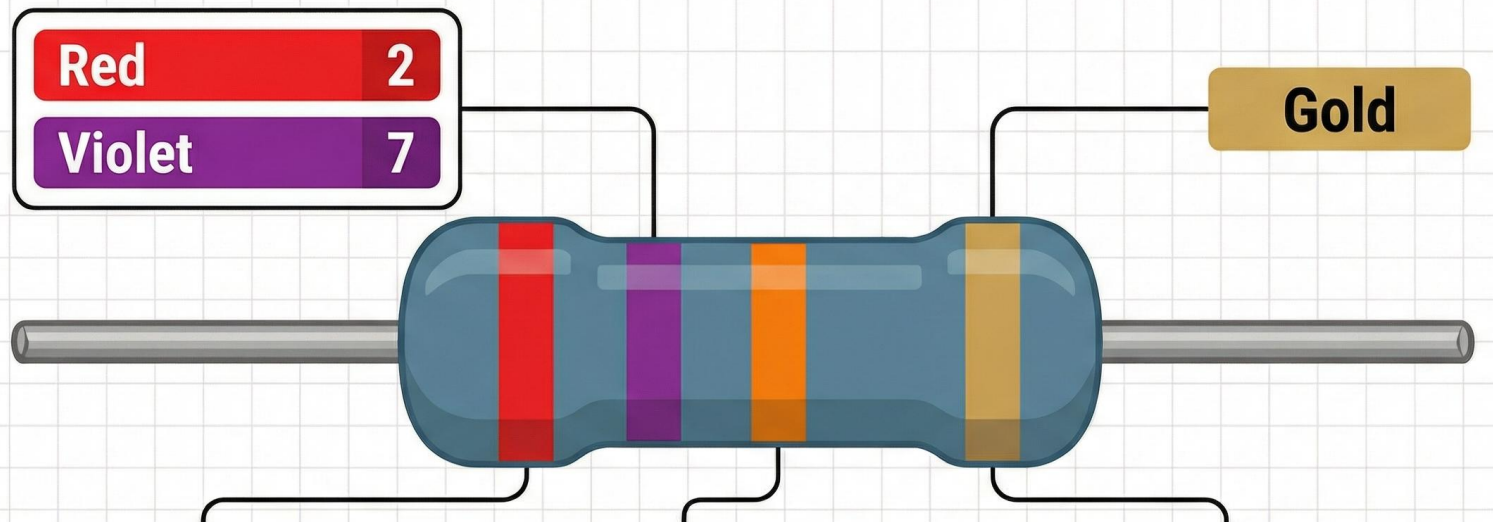


Thanks

Do you have any questions?



How to Read Resistor Color Codes



READ FIRST TWO DIGITS

Red (2) & Violet (7) 27

APPLY THE MULTIPLIER

Orange band = $\times 10^3$ (or $\times 1\text{k}\Omega$)

The third band tells you how many zeros to add (e.g., Red = $\times 10^2$, or add two zeros).

CHECK TOLERANCE BAND

Gold band = $\pm 5\%$ Accuracy

STANDARD COLOR SEQUENCE

Black (0)

Brown (1)

Red (2)

Orange (3)

Yellow (4)

Green (5)

Blue (6)

Violet (7)

Grey (8)

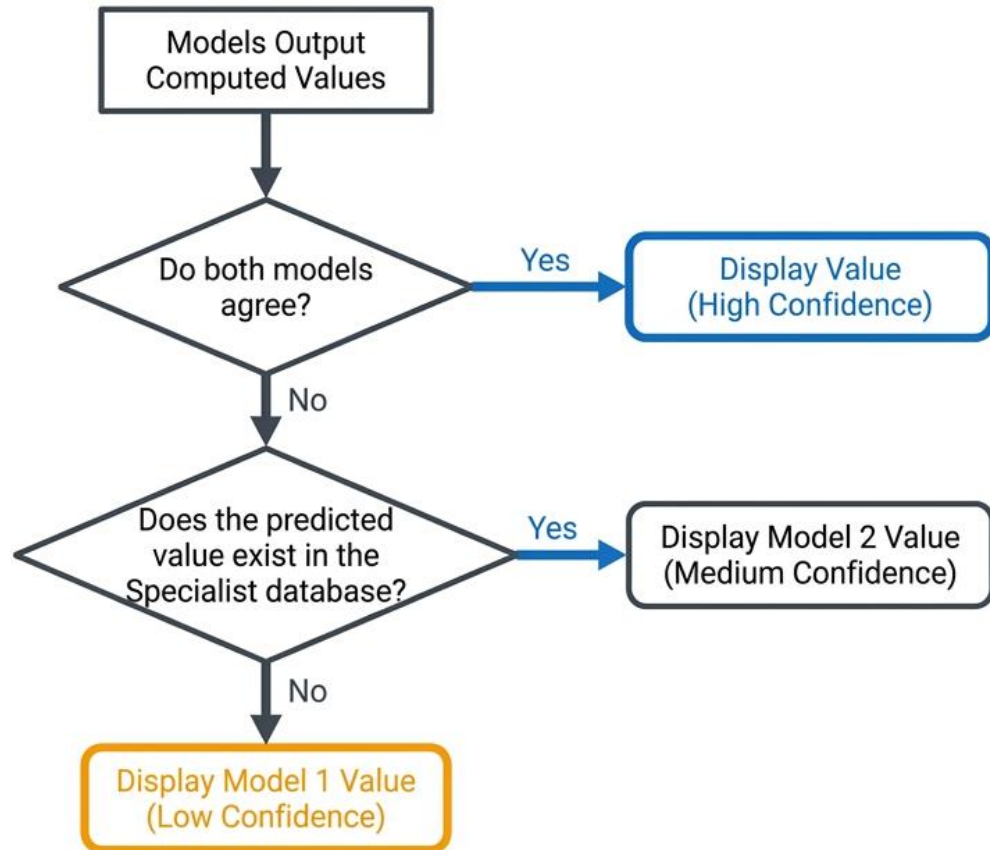
White (9)

Tolerance

Gold $\pm 5\%$

Silver $\pm 10\%$

Smart System Logic Resolves Output Conflicts



The system defaults to high confidence upon consensus, but systematically downgrades trust and defaults to the Specialist when conflicts arise.

Confusion Matrix Normalized

